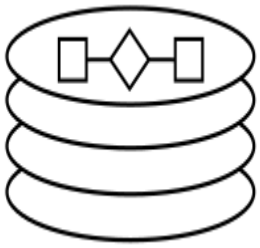
Sample Student

V0123456789

Group -1



CSC 370: Database Systems
Fall 2026, A01, A02
Data Modelling Practice Exam
Monday, 19 October 2026
13:05-14:20

Multiple Choice Knowledge Section. Each question is worth 3% and has one unique best answer. Record your response on the Crowdmark bubblesheet. Estimated time: 15 minutes.

Question 1: Which of the following best describes a reason for starting the construction of a database with conceptual design?

(a) Because you want to use an entity-relationship database management system (ERDBMS)
(b) Because you don't want to start with something formally precise when you're still just ideating
(c) Because you cannot build a relational database unless you have an entity-relationship diagram
(d) Because entity-relationship diagrams are always *normalised*
(e) Because it makes your high-level design independent of your choice of logical data model

Question 2: Which of the following terms refers to the *rows* of a relation?

(a) Schema (b) Tuple (c) Attribute (d) Instance (e) None of the above

Question 3: Which of the following statements best describes the importance of determining the multiplicity of a relationship?

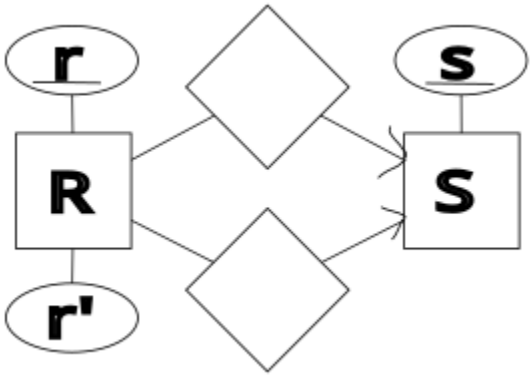
(a) It documents constraints on how entities can be related to each other
(b) It simplifies the conceptual design
(c) It is syntactically incorrect to not have any many-one or one-many relationships
(d) It ensures that hand-drawn ERDs are consistent with those generated by drawing tools in MySQL Workbench and Visio
(e) It makes it easier to determine functional dependencies later

Question 4: Which of the following is a key for relation R(A,B,C,D,E), given the three functional dependencies below?

$AB \rightarrow E$ $B \rightarrow C$ $C \rightarrow D$

(a) {A,B,C,D,E} (b) {A,B,C} (c) {A,B} (d) {B} (e) {A,C}

Question 5: Consider the following entity-relationship diagram (ERD). Assume there are n = |R| entities in entity set R and m = |S| entities in entity set S. What is the upper bound on how many different S entities can each r value be related to?



(a) 1 (b) 2 (c) n (d) m (e) 2m

Question 6: Which of the following are required in order for a relational algebra expression to represent a constraint?

(a) It must be possible to rearrange terms such that one side of the expression is the empty set
(b) It must involve more than one relational operator
(c) It must involve some sort of predicate using the selection operator, σ
(d) It must have an equality operator, =
(e) There must be one unique way to write the expression

Question 7: Which statement below best describes the following relational algebra constraint?

$$\pi_x(R) \setminus \pi_y(S) = \emptyset$$

(a) S.y is a foreign key referencing R.x
(b) S.y describes the set of possible values for R.x
(c) Each value in R.x can appear at most once in S.y
(d) Each value in R.x appears exactly once in S.y
(e) R.x and S.y are disjoint sets

Question 8: What is the result of the following SQL code after selecting a database to use?

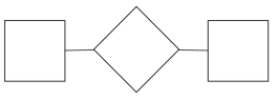
```SQL
CREATE TABLE `R`( `id` INT );
CREATE TABLE `S`( `id` INT, `r` INT );
ALTER TABLE `R` ADD COLUMN `s` INT;
DROP TABLE `S`;
```

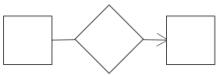
(a) There are two tables, each with two attributes
(b) There are two tables with different numbers of attributes
(c) There is one table with one attribute
(d) There is one table with two attributes
(e) Nothing. The code contains an error.

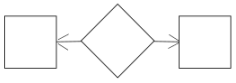
Question 9: Which of the following functional dependencies is the best fit for the following set of attributes: StackOverflowPosts(post_body, user_id, tag, timestamp, badge_name)

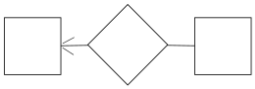
(a) user_id → post_body (b) post_body → user_id
(c) timestamp → badge_name user_id (d) timestamp → user_id post_body
(e) user_id → badge_name

Question 10: Which of the following entity-relationship diagrams (ERDs) contains a many-many relationship?

(a) 

(b) 

(c) 

(d) 

(e) More info is needed

Multiple Choice Skills Section. Each question is worth 5% and has one unique best answer. Record your response on the Crowdmark bubblesheet. Estimated time: 10 minutes.

Question 11: You are given relation R(A,B,C,D,E) and the functional dependencies below. Which of the following options would result from BCNF decomposition of R?

AB → E E → BC B → CD

(a) R(A, B, C, D, E)

(b) S(A,B,E)
T(B,C,E)
U(B,C,D)

(c) S(B,C,E)
T(A,D,E)

(d) S(B,C,D,E)
T(A,E)

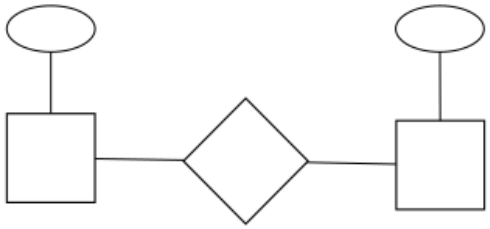
(e) S(B,C,D)
T(B,E)
U(A,E)

Question 12: Assume the three following things:

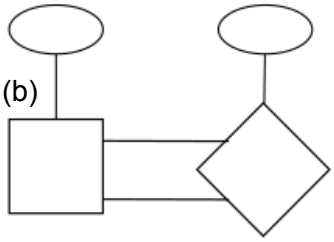
- you want to model a system with social networking features in which users can follow as many other users as they want, with no constraints on who they can follow;
- you also want to differentiate who started followed whom before your platform changed names when it was taken over by new management on 2022-10-08; and
- you want the simplest design possible that works, where "simple" means it leads to the fewest tables when implemented as a relational database.

Which ERD below best fulfills all three of these requirements?

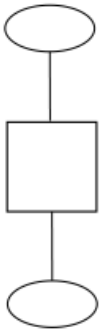
(a)



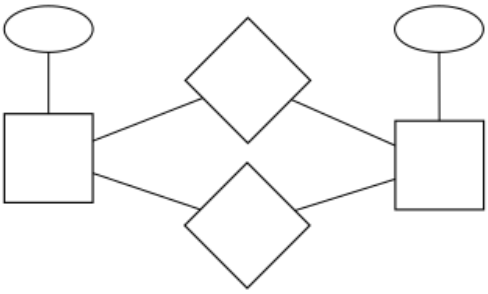
(b)



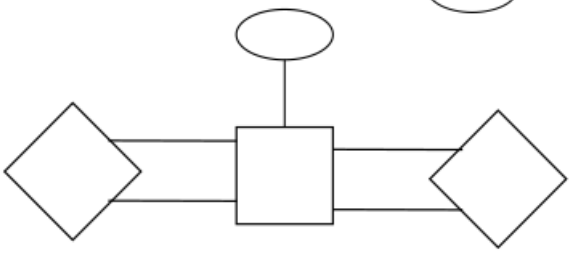
(c)



(d)



(e)



Question 13: Assume that you have a relation R(A,B,C,D). What is the closure of {A}, given the following functional dependencies: AB → AC B → BC A → B

(a) {} (b) {A} (c) {A,B} (d) {A,B,C} (e) {A,B,C,D}

Multiple Choice Integration Section. Each question is worth 5% and has one unique best answer. Record your response on the Crowdmark bubblesheet. Estimated time: 10 minutes.

Question 14: Assume that you have a relational database with four tables and two foreign keys. Consider that a third foreign key is added somewhere as a change to the design. Which of the following statements is *definitely* true?

(a) There is a many-to-many relationship in the corresponding ERD
(b) The new foreign key is added to a table that previously did not have one
(c) One of the entity sets in the original ERD must have been a weak entity set
(d) Every entity set is involved in at least one relationship in the new design
(e) None of the above

Question 15: Assume that you correctly convert an ERD into a relational schema as follows:

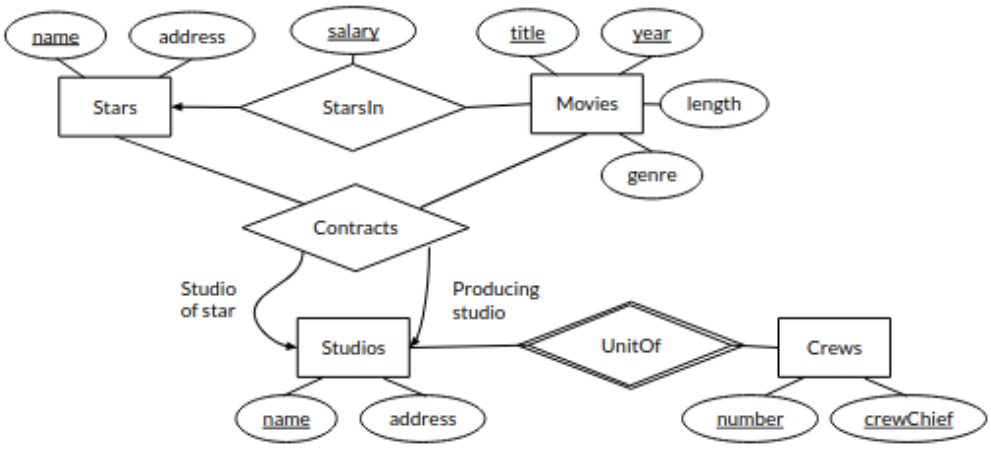
R(A,B, C) S(A, D, E) T(A, F, G) U(A, H, I)

If attribute names are unique, which of the following statements can be asserted with certainty about the original ERD?

(a) The schema was in BCNF
(b) There was only one entity set
(c) Attribute A is the only identifier
(d) S is a subset
(e) None of the above

Short Answer Project Section. The first three questions are worth 5%; the last two are worth 10%. Record your response directly underneath each question. Estimated time: 25 minutes.

Project Assets (e.g., ERD, SQL DDL code)



Question 16 (5%): In your team's ERD, there are two diamonds around the UnitOf relatic Assume that this is changed so that there is only one. Provide an example of an entity that would be valid in the new/changed design that would not have been valid in the one that you submitted.

Question 17 (5%): List all functional dependencies that are implied by your team's ERD and that involve attributes of the entity set Studios.

Question 18 (5%): New requirement: Consider that a Movie could have more than one genre. Illustrate how your ERD would change to support this requirement.

Question 19 (10%): Consider that salary is no longer underlined. Provide the new SQL DDL code to create all tables that are affected by this change.

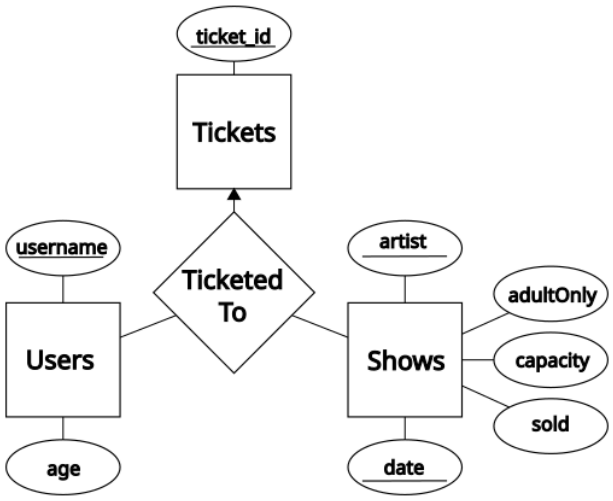
Question 20 (10%): With reference to the design criteria introduced in class (except completeness), evaluate in up to 3 sentences whether your team's ERD is better than an alternative in which Contracts is an entity set and there are binary relationships between Contracts and Stars, Movies, and Studios.

Design Section. Record your response directly underneath this question. You are evaluated on the quality of your explanation. Estimated time: 15 minutes.

Question 21 (10%): You are provided with the ERD below and the following set of requirements:

- users can obtain and purchase tickets to shows
- users have a unique username and must report their age, which cannot be negative
- an artist can only play one concert per day
- shows cannot be booked over-capacity
- some shows are adult-only
- when a payment goes through, the status of a ticket should change from 'booked' to 'paid' and the payment should be linked to the ticket

This design is poor. Illustrate how you would change the ERD and, with reference to the evaluation criteria introduced in class, justify (4 sentences) why your change is an improvement.



Also, I have the following check constraints set up on my tables:

- on age that it is non-negative
- on tuples in Shows that capacity ≥ sold

END OF EXAMINATION

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Name: _____ **Student ID Number:** _____

Instructions:

Please completely fill in the rectangle associated with your response. Example: ■ ☐B ☐C ☐D ☐E

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